

Project Name	Factorial Calculator App
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Created Date	10/01/25
Peer Review By	
Peer Review Date	

Test Id	Test case title	Test case description	Priority	Preconditions	Steps to execute	Test data	Expected results	Status	Actual results	Comments(if any)
TC_FCA_001	Verify the spelling of the page title on the browser tab.	Ensure that the page title displayed on the browser tab should be spelt correctly.	Low	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2. Hover on the browser tab in order to see the full page title.	none	The title should be Factorial.	Fail.	The title displayed is "Factorial!" instead of "Factorial".	
TC_FCA_002	Verify that there's a terms & conditions page.	Ensure that the Terms and Conditions hyperlink leads to the Terms and Conditions page when clicked.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the "Terms and Conditions" hyperlink at the bottom of the page. 3. Ensure that the clicked link is fully loaded or rendered.	none	The hyperlink should lead to the "Terms and Conditions" page.	Fail.	The "Terms and Conditions" hyperlink leads to the "Privacy" document page instead of the "Terms and Conditions" page.	
TC_FCA_003	Verify that there's a Privacy page.	Ensure that the Privacy hyperlink leads to the Privacy page when clicked.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the "Privacy" hyperlink at the bottom of the page. 3. Ensure that the clicked link is fully loaded or rendered.	none	The hyperlink should lead to the "Privacy" page.	Fail.	The "Privacy" hyperlink leads to the "Terms and Conditions" document page instead of the "Privacy" page.	
TC_FCA_004	Verify that there's an About Page	Ensure that the About hyperlink leads to the About page when clicked.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the "About" hyperlink at the bottom of the page. 3. Ensure that the clicked link is fully loaded or rendered.	none	The hyperlink should lead to the "About" page.	Pass.	The "About" hyperlink leads to the "About" page as expected.	
TC_FCA_005	Verify that the copyright message is correctly formatted.	Ensure that the copyright message exists and is correctly formatted.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Check the copyright message format.	none	The copyright message should only show or have the year 2025 once since the copyright only spans a single year.	Fail.	The copyright message shows a range of 2025-2025 instead of one single 2025.	
TC_FCA_006	Verify the spelling of the landing page heading.	Ensure that the spelling and the grammar of the H1 tag displayed on the landing page is correct.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Check the H1 tag text's spelling and grammar.	none	The H1 tag text should be properly spelt and follow the correct grammar.	Pass.	The h1 tag text is correctly spelt and grammatically correct.	
TC_FCA_007	Verify the spelling of the text box instruction.	Ensure that the spelling and the grammar of the instruction displayed inside the text box on the landing page is correct.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Check the spelling and the grammar of the instruction inside the text box.	none	The instruction inside the text box should be correctly spelt and should be grammatically correct.	Pass.	The instruction inside the text box is correctly spelt and grammatically correct.	I'd recommend the addition of "Please" just before "Enter" to improve user experience.
TC_FCA_008	Verify the spelling of the Calculate button.	Ensure that the spelling of the text on the button is correct.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Check the spelling of the text on the "Calculate!" button.	none	The spelling of the text on the "Calculate!" button should be correctly spelt.	Pass.	The text on the "Calculate!" button is correctly spelt.	
TC_FCA_009	Verify the functionality of the calculator with a positive number.	Ensure that the calculator can calculate the factorial of a positive integer correctly.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the positive integer from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	Positive_integer =12	The app should provide the correct answer of the factorial of 12 which is 479001600.	Pass.	The app provides the correct answer of the factorial of 12 which is 479001600.	
TC_FCA_010	Verify how the app handles alphanumeric input.	Ensure that the app communicates a message to the user to warn them that they should only use positive integers as inputs.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the alphanumeric input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	alphanumeric_input =1b2	The app should communicate a warning message to the user to "please enter an integer".	Pass.	The app correctly communicates a warning message to the user , the message is highlighted in red as well the the text box to show the user that their input is invalid.	I'd recommend a change in the error message being displayed or communicated from user to "please enter a positive integer " simply due to the fact that factorials only work with positive integers .
TC_FCA_011	Verify how the app handles decimal with remainders. inputs.	Ensure that the calculator app doesn't accept decimals with remainders as valid inputs and instead displays communication to the user to only use positive integers.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the decimal input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	decimal_input =1.5	The app should communicate or display a warning message to the user to "please enter an integer".	Pass.	The app correctly communicates a warning message to the user , the message is highlighted in red to show the user that their input is invalid.	I'd recommend a change in the error message being displayed or communicated from a warning message to the user , the message is highlighted in red to show the user that their input is invalid.
TC_FCA_012	Verify how the app handles negative integers as input.	Ensure that the calculator app doesn't accept negative integers as valid inputs and instead displays communication to the user to only use POSITIVE integers.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the negative integer input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	negative integer input = -5	The app should communicate or display a warning message to the user to "please enter an integer".	Fail.	There is no response or any communication from the App.	The app should communicate or display a warning message to the user to "please enter a positive integer " as factorials only work with positive integers .
TC_FCA_013	Verify how the app handles positive integers with white spaces in between.	Ensure that the app doesn't accept any white spaces in between positive integers used as input and instead displays communication to the user to only use valid inputs.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the invalid positive integer with white spaces from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	Invalid_positive_integer = 1 2	The app should communicate or display a warning message to the user to "please enter an integer".	Pass.	The app correctly communicates a warning message to the user , the message is highlighted in red as well the the text box to show the user that their input is invalid.	However I'd recommend that the message being communicated or displayed to the user be more specific in stating that "please enter a positive integer with no white spaces " since that was our input.
TC_FCA_014	Verify how the app handles integers with leading zeros.	Ensure that the app is able to calculate the factorials of any numbers with leading zeros correctly.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the leading zeros input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	leading_zeros_input =00000012	The app should provide the correct answer of the factorial of 12 which is 479001600.	Pass.	The app provides the correct answer of the factorial of 12 which is 479001600.	
TC_FCA_015	Verify how the app handles inputs with just white spaces.	Ensure that the app doesn't accept any white spaces as input and instead displays communication to the user to only use valid inputs.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the white spaces input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	white_spaces_input =	The app should communicate or display a warning message to the user to "please enter an integer".	Pass.	The app correctly communicates a warning message to the user , the message is highlighted in red as well the the text box to show the user that their input is invalid.	The app should communicate or display a warning message to the user to "please enter a positive integer " as factorials only work with positive integers .
TC_FCA_016	Verify how the app handles an input that exceeds the calculator app's data limit.	Ensure that the app can handle inputs that exceed the app's data limit gracefully.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the BVA_value input from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial.	BVA_value = 171	The app should handle the large value gracefully and display the answer as "infinity".	Pass.	The app handles the input gracefully and correctly displays the answer "infinity".	I'd recommend that the app properly lets the user know that it has reached it's data limit instead of just "infinity".
TC_FCA_017	Verify that invalid inputs trigger visual feedback styling.	Ensure that the calculator app correctly displays appropriate visual feedback when invalid inputs are provided.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the different inputs from Input_1 to Input_5 one at a time from the test data provided. 4. Click on the "Calculate!" button in order to calculate the factorial. 5. Verify the visual feedback. 6. Clear the input field. 7. Repeat steps 3-6 for each test input until Input_5.	Input_1 = 1 0 Input_2 = (two white spaces) Input_3 = 1.5 Input_4 = 1\$b Input_5= -12.5	The text box border as well the warning or response message should change from its current colour to a red colour.	Pass.	The text box border as well as the response message correctly changes colour from its current colour to a red colour as expected.	
TC_FCA_018	Verify that invalid inputs fail to trigger visual feedback for specific cases.	Ensure that the calculator app consistently displays the correct visual feedback for all invalid inputs and identify cases where it fails to do so.	Medium	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the different inputs from Input_1 to Input_3 one at a time from the test data provided. 4. Click on the "Calculate!" button to calculate the factorial. 5. Verify the visual feedback. 6. Clear the input field. 7. Repeat steps 3-6 for each test input until Input_3.	Input_1 = -12 Input_2 = -12.00 Input_3 = 12.00	The text box border as well the warning or response message should change from its current colour to a red colour.	Fail.	The text box border doesn't change colour to red as expected but the warning or response message is red.	
TC_FCA_019	Verify that a console message is printed upon a successful calculation.		Low	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the positive integer from the test data provided. 4. Open the browser developer tools(F12) . 5.Select the Console tab on the browser developer tools. 6. Click on the "Calculate!" button on the app to calculate the factorial. 7.Check the browser console for any messages.	Positive_integer =12	The should be a message reading "Hello! I am in the done part of the ajax call" being printed or displayed in the console logs.	Pass.	The app correctly prints or displays the message as expected.	
TC_FCA_020	Verify that invalid inputs fail to trigger a console message upon calculation.	Ensure that the calculator app doesn't print a console message for an invalid input.	Low	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the invalid_input from the test data provided. 4. Open the browser developer tools(F12) . 5.Select the Console tab on the browser developer tools. 6. Click on the "Calculate!" button on the app to calculate the factorial. 7.Check the browser console for any messages.	invalid_input = -12	The app shouldn't print or display any console message except for the input we initially fed it for calculation.	Pass.	The app rightfully doesn't print or display any console messages	
TC_FCA_021	Verify that an API call is successfully made when the input is valid.	Ensure that the app successfully makes an API call when a valid input is calculated.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the valid_input from the test data provided. 4. Open the browser developer tools(F12) . 5.Select the Network tab on the browser developer tools. 6. Click on the "Calculate!" button on the network tab, verify: -request URL matches factorial API endpoint. -API call method/type -API parameters(form-data). -API Status Code -API response.	valid_input = 12	Ensure that API call is successfully made by ensuring that our call has a status code of 200 & our response should have the correct answer which is 479001600.	Pass.	The API call is successfully made , we have our correct answer of 479001600, not forgetting that the status code of call is 200 meaning everything went well.	
TC_FCA_022	Verify that our API call becomes unsuccessful when an invalid input is used.	Ensure that the API call made when using an invalid input is unsuccessful.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the invalid_input from the test data provided. 4. Open the browser developer tools(F12) . 5.Select the Network tab on the browser developer tools. 6. Click on the "Calculate!" button on the app to calculate the factorial. 7.In network tab, verify: -request URL matches factorial API endpoint. -API call method/type -API parameters(form-data). -API Status Code -API response.	invalid_input = -12	Ensure that the API call is unsuccessful , has a status code of 500 and the response has no answer but an error message.	Pass.	The API call is unsuccessful, has a 500 status code and has no answer but an error message.	
TC_FCA_023	Verify how the app handles decimal remainders.	Ensure that the calculator app doesn't accept decimals without remainders as valid inputs and instead displays communication to the user to only use positive integers.	High	Ensure the landing page of the App is successfully loaded.	1.Open the app URL in order to load the landing page. 2.Click on the text box. 3.Enter the decimal without a remainder as input from the test data provided. 4.Click on the "Calculate!" button on the app to calculate the factorial.	decimal_input =1.0	The app should communicate or display a warning message to the user to "please enter an integer".	Fail.	There is no response or any communication from the App.	The app should communicate or display a warning message to the user to "please enter a positive integer " as factorials only work with positive integers .